

Analytical validation for quantum mechanic energy solver

The following document aims to determine the accuracy of the quantum mechanic energy solver by comparing the computational values to derived values of the Schrödinger's equation. The accuracy percent of the computational solver shall be calculated through the equation:

$$Accuracy = \left(1 - \frac{|derived\ value - computational\ value|}{|derived\ value|} \right) \times 100\%$$

The computational solver shall be evaluated against three distinct solutions of the Schrödinger equation, derived by three separate potential functions, $V(x)$.

However, due to the characteristics of the numerical method used, the accuracy of the calculated energy values varies between energy levels, as demonstrated analytically below. Hence the range and average of the accuracy percentages will be considered when evaluating whether the computational solver can be considered accurate for the first five energy levels.

Constant variables:

As the computational solver utilises the finite difference method, it is well known that as the number of data points increase, the accuracy of the generated values is increased. However, an increase in data points requires more computational power to solve. Hence, to ensure similar testing conditions for the three separate potential functions, the number of data points set on the computational solver will be 1000. Additionally as altering the maximum and minimum x of the graph also differ the accuracy, the range of the x values set on the computational solver will be between the maximum and minimum x shall be set to 10. Hence the strict spatial confinement for the computational trials will be set so that the range of the x values is 10 and the number of data points is set to 1000.

Additionally, as the mass (m) and reduced planck constant ($\hbar$) are set to 1 in the computational solver, when values are plugged into the general energy equation for each potential function, the mass and the reduced planck constant must also be set to 1. Other variables used within the potential functions chosen such as classical angular frequency (ω), and the magnitude of the delta potential function (a), will also be set to equal 1 to simplify the mathematics.

Test 1: Infinite Square Well

As seen under the “Derivations” branch, the derived general energy equation for the infinite square well solution to the Schrödinger equation is:

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}, \quad n = \{1, 2, 3, \dots\}, \quad L = x_{max}$$

Setting $L = 10$, $\hbar = 1$ and $m = 1$, the exact analytical energies for the ground state ($n = 1$) through the first four excited states ($n = 5$) are:

$$E_1 = 0.049348$$

$$E_2 = 0.197392$$

$$E_3 = 0.444132$$

$$E_4 = 0.789568$$

$$E_5 = 1.233701$$

When the following parameters are inputted into the computational solver:

$$x_{max} = 10$$

$$x_{min} = 0$$

$$Nx = 1000$$

$$V(x) = 0$$

Boundary condition = Dirichlet

The computational solver yields the following values:

$$E_1 = 0.049348$$

$$E_2 = 0.197391$$

$$E_3 = 0.444129$$

$$E_4 = 0.789558$$

$$E_5 = 1.233675$$

Hence, the accuracy percentage for each set of energy value gives:

$$A_{E_1} = 100\%,$$

$$A_{E_2} = 99.999493\%,$$

$$A_{E_3} = 99.999325\%,$$

$$A_{E_4} = 99.998733\%$$

$$A_{E_5} = 99.997893\%$$

Therefore, for the given strict spatial confinement parameters and the infinite square well potential function, the numerical results prove that the computational solver is accurate, as within the range of 5 separate energy levels, the accuracy percentage deviated 0.002107% whilst the average was 99.999089%.

Test 2: Quantum Harmonic Oscillator

As seen under the “Derivations” branch, the derived general energy equation for the quantum harmonic oscillator solution to the Schrödinger equation is:

$$E_n = \hbar\omega\left(\frac{1}{2} + n\right), \quad n = \{0,1,2, \dots\}$$

Setting $\hbar = 1$ and $\omega = 1$ the exact energies for the ground state ($n = 0$) through the first four excited states ($n = 5$) are:

$$E_0 = 0.5,$$

$$E_1 = 1.5$$

$$E_2 = 2.5$$

$$E_3 = 3.5$$

$$E_4 = 4.5$$

When the following parameters are inputted into the computational solver:

$$x_{max} = 5$$

$$x_{min} = -5$$

$$Nx = 1000$$

$$V(x) = \frac{1}{2}m\omega^2x^2$$

$$\omega = 1$$

Boundary condition = Dirichlet

The computational solver yields the following values:

$$E_1 = 0.499997$$

$$E_2 = 1.499984$$

$$E_3 = 2.499959$$

$$E_4 = 3.499923$$

$$E_5 = 4.499884$$

Hence, the accuracy percentage for each set of energy value gives:

$$A_{E_1} = 99.999400\%$$

$$A_{E_2} = 99.998933\%$$

$$A_{E_3} = 99.998360\%$$

$$A_{E_4} = 99.997800\%$$

$$A_{E_5} = 99.997422\%$$

The reason why the range of the values was set from -5 to 5 instead of the previous 0 to 10 is because for all the solutions to exist for this specific potential function, the function must be allowed to exist in both the positive and negative x-axis. However, to ensure that each trial has the same constant variables, the values of -5 to 5 have the same units of range as a range of 0-10, which ensures the spatial confinement for each experiment stays equivalent.

Given the numerical results, it is clear that the accuracy of the computational solver is highly accurate for the set spatial confinement and the specific potential function as between the 5 energy levels, the accuracy only deviated 0.001978% whilst the average accuracy was 99.998383%.

Test 3: Delta Potential Well centered at the Origin within an infinite square well

As seen under the “Derivations” branch, the Delta Potential well centered at the origin solution to the Schrödinger equation has three different energies, the bound state energy:

$$E_1 = -\frac{ma^2}{2\hbar^2}$$

The scattering odd parity states:

$$E_{n,even} = \frac{n^2 \pi^2 \hbar^2}{2mr^2}, \quad n = \{1,2,3, \dots\}, \quad r = \text{Maximum } x$$

The scattering even parity states:

$$\frac{am}{\hbar^2} \tan r \sqrt{\frac{2E_{n,odd}m}{\hbar^2}} = \sqrt{\frac{2E_{n,odd}m}{\hbar^2}}, \quad n = \{1,2,3, \dots\}, \quad r = \text{Maximum } x$$

In this case the even parity state cannot be manipulated into a general equation, and hence the equation above must be solved graphically to find the even parity states.

The reasoning behind the odd parity states being stated as $E_{n,even}$ and the even parity states being stated as $E_{n,odd}$ is because the ground state was stated as E_1 , instead of as E_0 hence in this case E_2 and E_4 are the odd parity states, despite normally being phrased as E_1 and E_3 due to the shift in indices, with the same applying to even parity states.

Setting $\hbar = 1$, $a = 1$, $m = 1$ and $L = 5$, the exact energies for the ground state ($n = 1$) through the first four excited states ($n = 5$) are:

$$\begin{aligned} E_1 &= -0.5 \\ E_2 &= 0.197392 \\ E_3 &= 0.287312 \\ E_4 &= 0.789568 \\ E_5 &= 1.051323 \end{aligned}$$

When the following parameters are inputted into the computational solver:

$$\begin{aligned} x_{max} &= 5 \\ x_{min} &= -5 \\ Nx &= 1000 \\ V(x) &= -\text{delta} \\ a &= 1 \\ \text{Boundary condition} &= \text{Dirichlet} \end{aligned}$$

The computational solver yields the following values:

$$E_1 = -0.499897$$

$$E_2 = 0.197389$$

$$E_3 = 0.287317$$

$$E_4 = 0.789550$$

$$E_5 = 1.051310$$

Hence, the accuracy percentage for each set of energy value gives:

$$A_{E_1} = 99.979400\%,$$

$$A_{E_2} = 99.998480\%,$$

$$A_{E_3} = 99.998259\%$$

$$A_{E_4} = 99.997720\%$$

$$A_{E_5} = 99.998763\%$$

The reason why the range of the values was set from -5 to 5 instead of the previous 0 to 10 is because for all the solutions to exist for this specific potential function, the function must be allowed to exist in both the positive and negative x-axis. However, to ensure that each trial has the same constant variables, the values of -5 to 5 have the same units of range as a range of 0-10, which ensures the spatial confinement for each experiment stays identical.

Once again, the range was set as -5 to 5 as both the positive and negative x-axis are required to generate all solutions, with the range being exactly 10 units so that the trial fits within the set spatial confinement.

Evaluating the numerical data reveals that the accuracy of the computational solver for the following potential function is accurate as between the 5 energy levels, the accuracy only deviated 0.019363% whilst the average accuracy was 99.9945244%.

Conclusion

Given the three different potential functions and the analytical results, it is reasonable to conclude that the computational solver was extremely accurate given the strict spatial confinement of a range of 10 units and 1000 data points for all three of the separate potential functions.

However, the results cannot establish accuracy for potential functions beyond those investigated, and hence further experimentation and analytical validation are required to completely validate the computational solver as a general computational solver.

Additionally, it is imperative to state that all three solutions derived assumed the Dirichlet boundary condition, hence no analytical evaluation has been conducted for the Neumann boundary condition, which is also a choice on the solver.

Lastly, it is important to note that the general trend between the three results is that the accuracy decreases slightly as energy level increases, this deviation though is not enough to impact the accuracy of the results above as the solver was coded to only solve the energy for the first 5 energy levels.